

# Biometry

**Today's agenda:**

Introductions,

Syllabus,

Why we need stats,

Reproducibility and Openness

# Introductions

Please take a moment to tell us about yourself:

**01**

## Your Name

Introduce yourself to the class, including what you prefer to be called and your pronouns if you'd like to share.

**02**

## Research

Describe your current research interests, what systems or questions you study, or what you hope to work on.

**03**

## Fun Fact

Tell us something interesting or unique about yourself—hobbies, favorite travel spots, pets, or hidden talents!

# Syllabus: basic stuff

Class Meeting Days: T, Th

Class Meeting Time: 2:00 – 3:50 pm

Class Meeting Location: DAV 265

Instructor: Dr. Brian Maitner

Office Location: DAV 226 (but I'm usually in URL 106, because it's warmer)

Office Hours: TBD - ?

Email: [bmaitner@usf.edu](mailto:bmaitner@usf.edu)

Up to date syllabus: [https://github.com/bmaitner/biometry\\_course](https://github.com/bmaitner/biometry_course)

# Course Github page

|                                                                                                                                                                                                                                                                      |                                                                  |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------|
|  <b>bmaintr</b> time and location update <span>b495611 · 50 minutes ago</span>  <b>117 Commits</b> |                                                                  |                |
|  R_scripts                                                                                                                                                                            | updated urls                                                     | 19 hours ago   |
|  assignments                                                                                                                                                                          | changing pipe to base R                                          | yesterday      |
|  data                                                                                                                                                                                 | removed larger marine organism data files, updated readm...      | 19 hours ago   |
|  example_R_commented                                                                                                                                                                  | changing pipe to base R                                          | yesterday      |
|  example_R_markdown                                                                                                                                                                   | changing pipe to base R                                          | yesterday      |
|  final                                                                                                                                                                                | final                                                            | 9 months ago   |
|  lectures                                                                                                                                                                             | notes on pending lecture updates                                 | 19 hours ago   |
|  midterm                                                                                                                                                                              | changing pipe to base R                                          | yesterday      |
|  quizzes                                                                                                                                                                              | q6                                                               | 8 months ago   |
|  renv                                                                                                                                                                                 | renv                                                             | 9 months ago   |
|  syllabus                                                                                                                                                                             | time and location update                                         | 50 minutes ago |
|  .Rprofile                                                                                                                                                                            | renv                                                             | 9 months ago   |
|  .gitignore                                                                                                                                                                           | gi update, removed old syllabus, switching to md file that re... | yesterday      |
|  LICENSE                                                                                                                                                                             | Initial commit                                                   | between        |

## Full of useful stuff

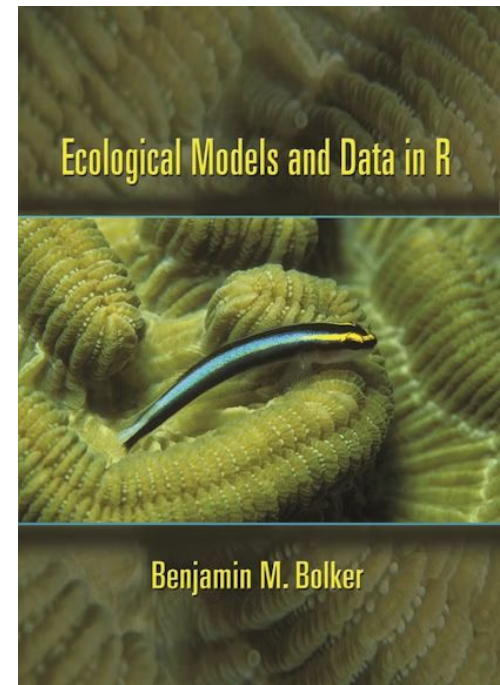
- Syllabus
- Lectures
- Previous quizzes

# Syllabus: Course structure

Flipped course: read before class, work during class

Assignments will focus on data of your choosing

| Graded Item                      | Percent of Final Grade |
|----------------------------------|------------------------|
| In-class participation           | 20%                    |
| In-class quizzes                 | 20%                    |
| Assignments (4x)                 | 20%                    |
| Midterm (take-home)              | 20%                    |
| Final (presentation + take-home) | 20%                    |



Digital copies available  
through the library for free

# Course Expectations



## Participate!

Ask questions. Share ideas, experiences, and knowledge!



## Be Respectful.

Create a welcoming environment for everyone to learn, collaborate, and grow together.

# Recommendations



**Read the  
syllabus**



**Read the book**



**Take notes on  
paper**



**Plan ahead**



**Bring a laptop**

**Questions so far?**



# What will you get out of this class?

In general:



## Practical Application

Ability to do common analyses  
and visualizations



## Future Growth

Ability to go further on your own

# What will you get out of this class?

More specifically:



## DATA & VISUALIZATION

- Load data into R & perform common data-wrangling
- Create plots using base R & ggplot2



## STATS & ANALYSIS

- Understand key statistical distributions and applications
- Select correct analyses for ecological datasets



## PROBLEM SOLVING

- Troubleshoot & debug R code effectively
- Approach novel coding challenges with confidence
- Apply new programming skills directly to your own work

# Learning Stats

People learn differently and stats can be non-intuitive



## THE BOOK

Exposes you to verbal and mathematical descriptions of key statistical concepts.



## IN CLASS

Hands-on coding and interactive visualizations help build intuitive understanding.



## THE GOAL

Providing multiple teaching methods ensures that at least one appeals to everyone.

# Learning Stats



## What to Avoid

**Don't focus on  
memorization**



## What to Focus On

**DO focus on practice and  
understanding**

# A Note on AI

Establishing core fundamentals before leveraging artificial intelligence



## TIMING

We'll cover AI later in the semester, once the foundations are established.



## APPROACH

Strongly recommend that you start out without it to truly grasp the mechanics.



## REQUIREMENT

You need reasonable fundamentals to be able to effectively and safely use AI.

**Questions so far?**

# Why do we need statistics?

# Why do we need statistics?

## 01 . COGNITIVE LIMITS

### Imperfect Minds

We must overcome human biases, logical fallacies, and cognitive limitations.

## 02 . MEASUREMENT

### Effect Sizes

Accurately quantifying the strength and magnitude of different relationships.

## 03 . VALIDATION

### Hypothesis Testing

Systematically validating assumptions against empirical data.

## 04 . FORECASTING

### Making Predictions

Building robust models to anticipate future outcomes and trends.

## 05 . CLARITY

### Visualizations

Making complex numerical relationships instantly clear and intuitive.

## 06 . ETC

### Lots of Other Stuff

From optimizing systems to driving data-informed strategic decisions.



# What is R and why is it useful?

## THE CORE IDEA

### Scripted Language

Type **commands** instead of clicking buttons. This fundamental shift unlocks precision and power in data analysis.

## REPRODUCIBILITY

### Easily Re-run & Revise Analyses

Modify your inputs or parameters and re-execute entire workflows instantly.

## EFFICIENCY

### Re-use Code Across Projects

Build on top of previous work by recycling robust scripts for new datasets.

## DOCUMENTATION

### Code as a Methods Section

Your codebase serves as a transparent, step-by-step record of your methodology.

# What is R and why is it useful?

## 01 . COST-EFFECTIVE

### **R is Free to Use**

Anyone can download and use R without licensing costs, lowering the barrier to entry.

## 02 . EXTENSIBILITY

### **User-Contributed Packages**

A massive ecosystem of user-developed packages is contributed for free, extending its capabilities.

## 03 . VERSATILITY

### **Incredible Flexibility**

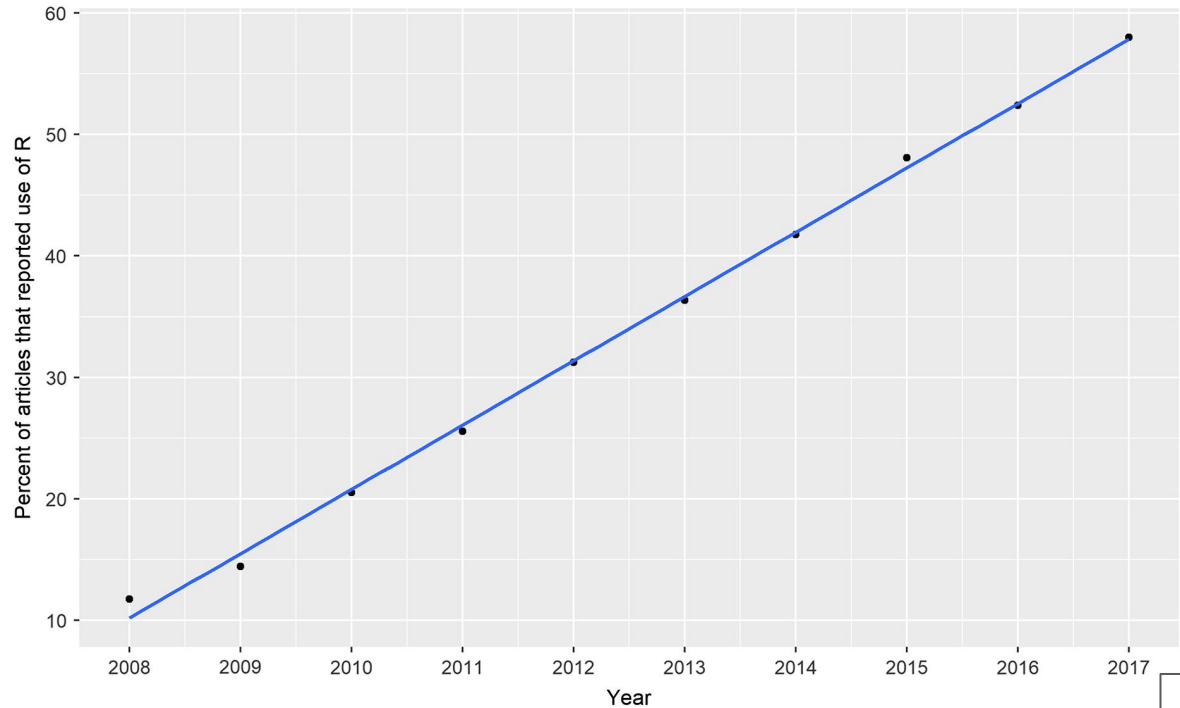
The vast range of libraries allows researchers to handle virtually any quantitative analysis task.

## 04 . COMMUNITY

### **Popular & Supported**

An active, global developer and user base provides vast documentation and peer-to-peer help.

# What is R and why is it useful?



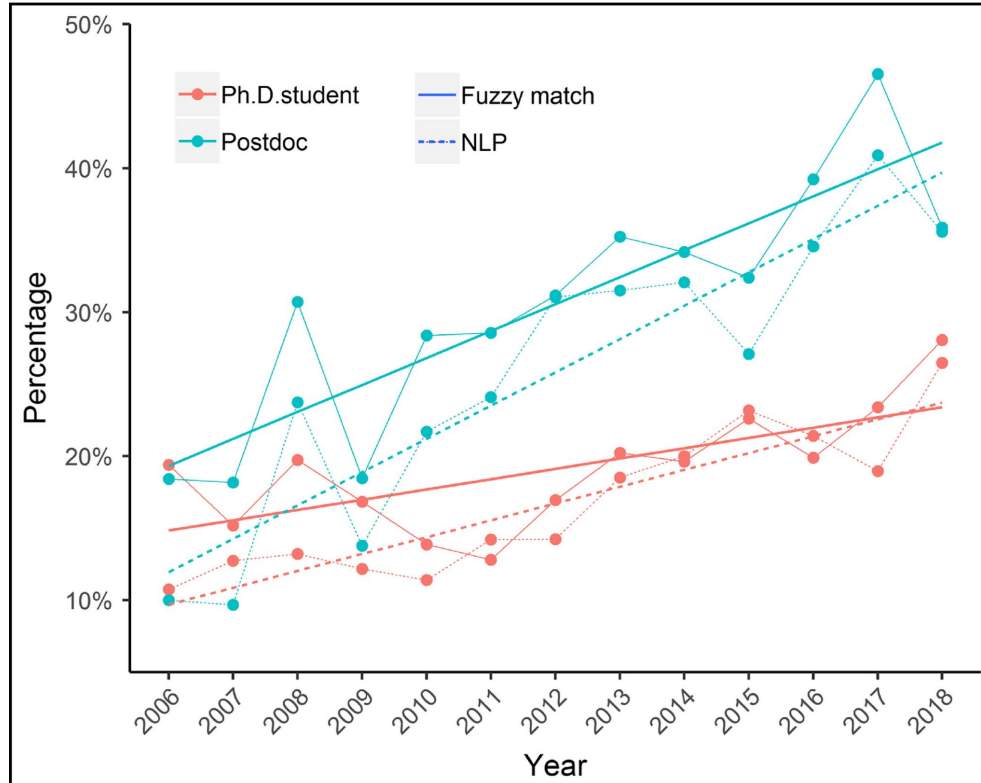
esa

ECOSPHERE

Evaluating the popularity of R in ecology

JIANGSHAN LAI<sup>1,2</sup>, † CHRISTOPHER J. LORTIE<sup>3,4</sup>, ROBERT A. MUENCHEN<sup>5</sup>, JIAN YANG<sup>6</sup> AND KEPING MA<sup>1</sup>

# What is R and why is it useful?



# Transparency & Reproducibility

Science needs to be open, clear, and verifiable.



## Transparency

What was done is clear



## Reproducibility

Can be re-done with the same results



**R helps with both of these!**

# Transparency and Reproducibility

## 2 MATERIALS AND METHODS

### 2.1 Data collection

#### 2.1.1 List of ecology and evolution publications citing R

To generate a list of papers in ecology and evolution that likely made use of the R programming language (R Core Team, 2023), we performed a query on the Scopus database (<https://www.scopus.com>) using the `scopus` R package (Muschelli, 2019). We searched Scopus (performed August 19, 2023) for peer-reviewed journal articles that: (1) included the words "ecology" or "evolution" in an "all fields" search (which searches article titles, keywords, abstracts, and journal titles); (2) were published in journals within the subject area "agriculture and biological sciences"; (3) were published after January 1, 2010; (4) were written in English (as this is currently the dominant language of publication in ecology and evolution; Mänttinen et al., 2010); and (5) included a citation of R in their reference list.

#### 2.1.2 Checking for code and data availability

We manually evaluated a randomly chosen subset of the publications on our overall list. We selected a total of 1001 papers, evenly distributed across the time period (77 year  $\times$  13 years). Papers that cited R but did not use it (or were unclear on whether they used it;  $n = 3$ ) were discarded and replaced by a randomly selected paper from the same year. For each publication in this subset, we manually identified whether the publication shared any R code, either as supplementary information, or via a link (e.g., to a GitHub repository). For each paper, we (i) checked for the presence of code in supplemental material, (ii) skimmed publications for code and data availability statements, (iii) searched through publications for terms associated with code (i.e., "code", "supplement", "appendix", "R", "script", "GitHub"), and (iv) searched publications for URLs. Papers were scored with a binary variable indicating whether they shared R code or not. We did not distinguish between publications which shared sufficient code for reproduction and those which did not. We also did not attempt to return the code or assess its reproducibility, and only recorded the presence of any code, even if it was incomplete. Where code was included, we recorded the license the code was provided under, or lack thereof. We also assessed whether publications were open access and whether they shared open data in order to understand the importance of open code relative to these other open-access components. Open access information was provided by the `scopus` R package (Muschelli, 2019). Open data was scored as a binary variable indicating whether the authors shared the full set of raw data underlying the analyses or not. To control for differences in citation rates among journals, we downloaded impact factor information using the `scholar` R package (Körner, 2016) on June 16, 2023. To estimate the proportion of publications which use R but do not properly cite it, we screened 130 randomly selected publications evenly distributed across the time period. These publications were selected using identical criteria to the publications that did cite R, except that they did not include R in their list of references.

#### 2.2 Checking for code citations

Where code was shared in a citable location such as a DOI or URL ( $n = 33$ ), we assessed whether the code itself was cited by querying the Scopus database for the URL (and DOI, where appropriate) using the `scopus` R package (Muschelli, 2019). Publications where code was shared in appendices or supplementary information ( $n = 22$ ) were excluded, as there was no way of distinguishing citations of the code with citations of the publication itself.

### 2.3 Analyses

All analyses were conducted in R version 4.3.0 (R Core Team, 2023). All R scripts underlying these analyses are available at: [https://github.com/bmaitner/R\\_citations](https://github.com/bmaitner/R_citations) and via Zenodo (Maitner & Lei, 2024). For data processing, we used the R packages `stringr` version 0.9.12 (van der Lee, 2014), `tidyr` version 2.0.0 (Wickham et al., 2018), `ggplot2` version 2.1.1 (D'Agostino McGowan & Bryan, 2023), and `googlehtest` version 1.1.1 (Bryan, 2023). For analyses, the packages `binde` version 1.0.25.1 (Baker & R Development Core Team, 2023), `binde` version 1.4.4 (Zhang, 2022), `MAIn` version 1.4.7.5 (Barton, 2023), `scopus` version 0.6.6 (Muschelli, 2019), `scholar` version 2.6 (Zhang, 2018), `scholar` version 0.2.4 (Körner, 2016), and `stats` version 4.3.0 (R Core Team, 2023). For plotting, the packages `ggplot2` version 3.4.4 (Wickham, 2016), `ggmargins` version 0.5.5 (Alpha), and `ggthemes` version 0.7.8 to identify all R packages used.

#### 2.3.1 Proportion of papers sharing code over time

We tested for a trend in code-sharing over time by modeling code sharing (binary, `yes/no`) as a function of the year (relative to 2010) using a generalized linear model. Modeling was performed using the function `glm` in the `stats` R package (R Core Team, 2023) with a binomial error distribution. We similarly tested for temporal trends in two other open-science components, open access publication (binary) and open data (binary). We also tested whether open access or open data papers were disproportionately likely to share code using `chi.square.test` via the `chisq.test` function in the `stats` R package (R Core Team, 2023).

#### 2.3.2 Impact of code sharing on citations

We additionally modeled the relationship between code sharing and citation count using generalized linear models in R. We modeled the dependent variable (cumulative number of citations of each article by 2022) using a Poisson distribution, which models the number of independent events occurring within a period of time (Baker, 2008). In addition to the predictor variable for code sharing (binary, `yes/no`), we included other variables that were hypothesized to influence citation count. Data sharing (binary, `yes/no`) may increase citation counts as readers may cite papers at data sources (Christensen et al., 2019; Priesner et al., 2007). Open access (binary, `yes/no`) may also increase citation counts by reaching a broader set of readers (Fang et al., 2017). Publications accumulate citations over time, and so citation count should increase with publication age (continuous, 1–13 years). Finally, publications in higher impact journals may be more likely to be read and cited, and hence, journal impact factor (continuous) may also influence citation counts. In addition to main effects, we considered two classes of interactions: (1) interactions between publication age and other main effects, which are appropriate if a main effect modifies the rate at which a publication accumulates citations over time; and (2) interactions between open access criteria (i.e., open access, open code, and open data), which are appropriate if there are synergistic effects of meeting multiple open-access criteria. We compared 11 models (including one null model that differed in complexity and that represented different hypotheses regarding the factors that influence citations (Table 1). Continuous variables were scaled and centered. Overall model pseudo  $R^2$  for the best-performing model was calculated using the function `r.squaredGLMM` in the `mgc` package (Zhang, 2018).

TABLE 1. Candidate models of citation count.

| ID | Models                      | #F | AIC  |
|----|-----------------------------|----|------|
| 1  | Citations ~                 | 13 | 0.0  |
|    | Impact factor ~ Age ~       |    |      |
|    | Code shared ~ Age ~         |    |      |
|    | Open access ~ Age ~         |    |      |
|    | Data shared ~ Age ~         |    |      |
|    | Data shared ~ Code shared ~ |    |      |
|    | Code shared ~ Open access ~ |    |      |
|    | Open access ~ Data shared   |    |      |
| 2  | Citations ~                 | 9  | 8343 |
|    | Impact factor ~ Age ~       |    |      |
|    | Code shared ~ Age ~         |    |      |
|    | Data shared ~ Age ~         |    |      |
|    | Data shared ~ Code shared   |    |      |

## Code sharing in ecology and evolution increases citation rates but remains uncommon

Brian Maitner<sup>1,2</sup> | Paul Efrén Santos Andrade<sup>3</sup> | Luna Lei<sup>4</sup> | Jamie Kass<sup>5</sup> | Hannah L. Owens<sup>6,7,8</sup> | George C. G. Barbosa<sup>9</sup> | Brad Boyle<sup>10</sup> | Matiss Castorena<sup>10</sup> | Brian J. Enquist<sup>10,11</sup> | Xiao Feng<sup>12</sup> | Daniel S. Park<sup>13,14</sup> | Andrea Paz<sup>15,16</sup> | Gonzalo Pinilla-Buitrago<sup>17</sup> | Cory Merow<sup>18</sup> | Adam Wilson<sup>2</sup>

# Transparency and Reproducibility

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Where code was shared in a citable location such as a DOI or URL ( $n = 33$ ), we assessed whether the code itself was cited by querying the Scopus database for the URL (and DOI, where appropriate) using the rscopus R package (Muschelli, 2019). Publications where code was shared in appendices or supplementary information ( $n = 22$ ) were excluded, as there was no way of distinguishing citations of the code with citations of the publication itself.

### 2.3 Analyses

All analyses were conducted in R version 4.3.0 (R Core Team, 2023). All R scripts underlying these analyses are available at: [https://github.com/bmaitner/R\\_citations](https://github.com/bmaitner/R_citations) and via Zenodo (Maitner & Lei, 2024). For data processing, we used the R packages singlr (version 0.9.12 (van der Lee, 2014), rstan version 2.20 (Wickham et al., 2019), ggplot2 (version 2.1.1 (D'Agostino McGowan & Bryan, 2023), and googlescholar (version 1.1.1 (Bryan, 2023)) for analyses, the packages binde (version 1.0.25.1 (Baker & R Development Core Team, 2023)), DfM (version 1.4.4 (Zhang, 2023)), Mafit (version 1.47.5 (Baron, 2023)), rscopus (version 0.6.6 (Muschelli, 2019), m7 (version 2.6 (Zhang, 2018)), scholar (version 0.2.4 (Körner et al., 2016)), and rstan version 4.3.0 (R Core Team, 2023)) for plotting, the packages ggplot2 (version 3.4.4 (Wickham, 2016)), ggvis (version 0.5.5 (Aphalo, 2023)), and qversion (version 0.7.8 to identify all R packages used).

#### 2.3.1 Proportion of papers sharing code over time

We tested for a trend in code-sharing over time by modeling code sharing (binary, yes/no) as a function of the year (relative to 2010) using a generalized linear model. Modeling was performed using the function glm in the stats R package (R Core Team, 2023) with a binomial error distribution. We similarly tested for temporal trends in two other open-science components, open access publication (binary) and open data (binary). We also tested whether open access or open data papers were disproportionately likely to share code using chi-square tests via the chi2test function in the stats R package (R Core Team, 2023).

#### 2.3.2 Impact of code sharing on citations

We additionally modeled the relationship between code sharing and citation count using generalized linear models in R. We modeled the dependent variable (cumulative number of citations of each article by 2022) using a Poisson distribution, which models the number of independent events occurring within a period of time (Baker, 2008). In addition to the predictor variable for code sharing (binary, yes/no), we included other variables that were hypothesized to influence citation count. Data sharing (binary, yes/no) may increase citation counts as readers may cite papers as data sources (Christensen et al., 2019; Priesner et al., 2007). Open access (binary, yes/no) may also increase citation counts by reaching a broader set of readers (Ting et al., 2013). Publications accumulate citations over time, and so citation count should increase with publication age (continuous, 1–13 years). Finally, publications in higher impact journals may be more likely to be read and cited, and hence, journal impact factor (continuous, 0–13) may be positively associated with citation count. In addition to main effects, we considered two classes of interactions: (1) interactions between publication age and other main effects, which are appropriate if a main effect modifies the rate at which a publication accumulates citations over time; and (2) interactions between open-science criteria (i.e., open access, open code, and open data), which are appropriate if there are synergistic effects of meeting multiple open-science criteria. We compared 11 models (including one null model that differed in complexity and that represented different hypotheses regarding the factors that influence citations (Table 1). Continuous variables were scaled and centered. Overall model pseudo  $R^2$  for the best-performing model was calculated using the function rsquaredGLM in the r2 package (Zhang, 2018).

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| ID | Models                      | #F | AIC   |
|----|-----------------------------|----|-------|
| 1  | Citations ~                 | 13 | 8.0   |
|    | Impact factor ~ Age ~       |    |       |
|    | Code shared ~ Age ~         |    |       |
|    | Open access ~ Age ~         |    |       |
|    | Data shared ~ Age ~         |    |       |
|    | Data shared ~ Code shared ~ |    |       |
|    | Code shared ~ Open access ~ |    |       |
|    | Open access ~ Data shared ~ |    |       |
| 2  | Citations ~                 | 9  | 834.3 |
|    | Impact factor ~ Age ~       |    |       |
|    | Code shared ~ Age ~         |    |       |
|    | Data shared ~ Age ~         |    |       |
|    | Data shared ~ Code shared ~ |    |       |

github.com/bmaitner/R\_citations/tree/v1.0

R\_citations Public

Go to file

Code

About

No description, website, or topics provided.

Readme

CC0-1.0 license

Activity

3 stars

2 watching

1 fork

Releases 3

Revision release Latest on Apr 28, 2024

+ 2 releases

Packages

No packages published

Publish your first package

Contributors 3

Overview

# Transparency and Reproducibility

Note: many journals and funders are now requiring code to be published



**Questions on any of this?**

## **Next time:**

Before class: read 1.1 - 1.3

During class:

- Discuss 1.1-1.3
- Installing and setting up R, RStudio, and Github (if needed)